

Weekly Progress Report - Week 4

Improve performance of existing online trackers for small object detection [UAV Videos]

Group: The Convolutioners

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Outline of Performed Tasks:

- Explored and compared multiple YOLO model variants for small object detection on UAV footage
- Fine-tuned the best-performing YOLO model on the VisDrone training set
- Evaluated all YOLO variants on the VisDrone 2019 validation set
- Implemented the SAHI (Slicing Aided Hyper Inference) tiling strategy on the selected YOLO model
- Compared BoT-SORT tracking performance with and without the SAHI tiling strategy
- Conducted an initial baseline evaluation of the Siamese network within the tracking pipeline

YOLO Model Exploration and Fine-Tuning

This week began with a systematic exploration of available YOLO model variants, including YOLOv8n and YOLO11, to identify the most suitable detector for the VisDrone dataset. Each model was evaluated on its ability to detect small and densely packed objects characteristic of UAV aerial footage.

A key limitation observed across all baseline YOLO models was that full VisDrone frames are significantly downsampled during inference, causing small objects to become effectively undetectable after resizing. Fine-tuning was performed on the VisDrone-MOT training sequences by converting VisDrone annotations into YOLO format, where each object’s class ID and bounding box are normalised to image dimensions as centre coordinates (c_x, c_y, w, h) . Annotations marked as ignored regions and bounding boxes with invalid dimensions were filtered out prior to training. YOLO11 was identified as the strongest performer and selected as the detection backbone for all subsequent experiments.

SAHI Tiling Strategy

To address the fundamental resolution bottleneck described above, the SAHI (Slicing Aided Hyper Inference) framework was integrated with the fine-tuned YOLO11 model. Rather than feeding a full downsampled frame to the detector, SAHI subdivides each input frame into overlapping 640×640 crops and runs inference independently on each tile. Predictions across all tiles are then merged back into the original image coordinate space using Non-Maximum Suppression (NMS) to resolve duplicate detections from overlapping regions.

This approach preserves the native resolution of small objects during inference, which is critical for the VisDrone dataset where pedestrians and vehicles may occupy as few as 10–20 pixels in a full-resolution frame.

BoT-SORT: Tiling vs. No-Tiling Comparison

A direct comparison of BoT-SORT tracking performance was conducted between two detection configurations:

- **Plain YOLO11 + BoT-SORT:** Standard full-frame inference without tiling
- **SAHI + YOLO11 + BoT-SORT:** Tiled inference with the SAHI framework

Initial results indicate that while the SAHI pipeline improves recall and reduces missed detections by recovering small objects from individual tiles, it also introduces a significant increase in false positives due to overlapping tile boundaries. This false-positive surge directly penalises MOTA, which is linear in false positive count. These findings motivate further investigation into improved NMS post-processing for the tiled pipeline.

Initial Siamese Network Baseline Evaluation

Following the Siamese network implementation from the prior week, an initial baseline evaluation was conducted by passing detection crops through the network within the tracking loop. The model was observed to converge quickly, with recall reaching 1.0 early in training, indicating the decision boundary is initially permissive. Embedding distance statistics confirmed that same-object distances and different-object distances are separating progressively across epochs, validating the contrastive training setup.

Table 1: Epoch-wise Training Performance of the Siamese Network (Initial Run)

Epoch	Loss	Acc	Prec	Recall	F1
1	2.666	0.664	0.603	0.972	0.744
2	0.216	0.677	0.607	0.999	0.755
3	0.139	0.717	0.635	1.000	0.777
4	0.098	0.791	0.703	1.000	0.826
5	0.083	0.771	0.687	1.000	0.814

The slight decline in epoch 5 metrics relative to epoch 4 suggests the onset of mild overfitting, motivating a more systematic hyperparameter search in the following week.

Tentative List of Tasks for Next Week:

1. Conduct a structured grid sweep across Siamese network hyperparameters including backbone architecture, embedding dimension, learning rate, and contrastive margin
2. Explore different similarity threshold values (τ) and their effect on precision, recall, F1, and accuracy
3. Evaluate optimised Siamese network performance on the held-out validation crop set
4. Perform Optuna Bayesian Optimisation to refine hyperparameters in the Focal-BCE training regime
5. Compile best-performing parameter configuration for tracker integration